Multi-Signal Housing Price Prediction

XGBoost Structured Model · RentCast AVM · CLIP Condition Scoring

Brantson Bui, Hanru Li
Machine Learning · April 2026

Related Works

Classical Approach (Hedonic Pricing / Linear Regression):Early work modeled housing prices as a linear combination of features such as square footage, number of rooms, and location. A well-known example is the Boston Housing Dataset, which demonstrated how statistical models can capture basic relationships but struggle with nonlinear patterns.

Modern Approach (Tree-Based Machine Learning):More recent studies use advanced Machine Learning models such as Random Forest and Gradient Boosting (e.g., XGBoost). These methods outperform linear models by capturing complex interactions between features and handling large, real-world datasets like Zillow Housing Data.

The Problem

Why is housing price estimation hard?

- Prices vary 10–100× across zip codes in the same metro area
- Structural features (sqft, beds) explain only ~28% of price variance alone
- Premium markets are sparse in training data — models underfit above $600k
- No single signal is reliable: AVM APIs miss luxury comps, Zillow requires exact addresses
- Visual condition (renovated vs dated) can shift value by 10–15% yet isn't captured in tabular data

$47T US residential real estate value

28% Price variance from sqft + beds alone

3 independent signals in our pipeline

Dataset

2.2M
homes
across 54 states/territories

40+
features per home
(structural, geographic, amenities)

3
data sources
merged into unified training set

Data Sources

Source	Homes	Coverage	Key Fields
Kaggle Structured Dataset	~2.2M	54 US states/terrirtories	bed, bath, sqft, city, state, zip, price
Zillow Scraped (non-CA)	~4,800	20+ states	Address, price, year built, property type
Realty In US API (photos)	89	39 states	Interior photos → CLIP calibration images

Prediction Pipeline

1

Structured Model

XGBoost on 40+ features (always)

2

AVM Blend

RentCast property estimate (w/ address)

3

Market Anchor

Zip price/sqft × sqft (w/ zip)

4

CLIP Condition

ViT-B/32 visual scoring (w/ photos)

Final Price Estimate

Signals blended with geography-aware weights + outlier detection

Outlier detection fires when AVM + market diverge from structured by >42% log-space → premium boost applied

XGBoost Structured Model

Feature Engineering (40+ features)

Geographic

City & zip target encoding, state ordinal, latitude/longitude proxy via encoding

Structural

sqft, bed/bath, bed-bath ratio, sqft/bed, $sqft^2$, is_large_house flag

Property

year_built → house_age, decade_bucket, property type one-hot (5 categories)

Lot & Extras

Acre lot (log), HOA fee (log), has_pool, garage spaces, stories

Training & Performance

Algorithm:	XGBoost (n_estimators=5000, early stopping)
Target:	log(1 + price) → expm1 to recover price
Split:	80% train / 20% hold-out (random_state=42)
GPU:	CUDA-accelerated training (~4–6 min)
Encoding:	Target Encoder for city + zip (CV=5, smooth=auto)

0.494
Val RMSE (log-price)

0.336
Val MAE (log-price)

26.1%
Premium MAPE >$600k

Multi-Signal Blending

RentCast AVM

- Property-level AVM estimate
- Blended 10–60% by confidence
- High confidence → up to 60% weight
- Divergence scaling: extra weight when
- AVM diverges from structured > 0.3 log

Zip Market Stats

- Price-per-sqft × home sqft
- Independent of listing-specific noise
- Added as 3rd signal when AVM present
- Corroboration check: ±0.30 log-space
- No API key required — fully free

Outlier Detection + Premium Boost

- Fires when AVM + market both diverge from structured by >42% log-space
- AND agree with each other within 0.30 log-space
- Structured weight drops to 5% of blend
- Premium boost up to +30% for non-CA markets

Geography-aware weights: CA uses conservative AVM blend (10-40%); non-CA uses higher weights (25-60%) — less structured training richness outside CA

CLIP Zero-Shot Condition Scoring

How It Works

- Load CLIP ViT-B/32 (400M image-text pairs)
- Compute cosine similarity: image vs 4 positive prompts (modern, renovated, high-end) and 4 negative prompts (dated, worn, deferred maintenance)
- condition_score = mean(positive sims) − mean(negative sims)
- Apply calibration: log-price adj = slope × score + intercept
- Adjust final blend by ±15% (±0.15 log) maximum
- Supports up to 5 photos — scores averaged equally

Calibration Results (15563 images, 39 states)

Pearson corr.	**+0.087**	*Directionally correct*
R^2 (linear)	**−2.64**	*Fallback mode active*
Adj. ceiling	**±15%**	*Max log-price shift*

Semantic Prompts

POSITIVE (modern / well-maintained)

"a modern home interior with updated finishes and contemporary design"

"a renovated kitchen with new appliances, granite countertops and fresh paint"

"a newly built house with clean modern bathrooms and hardwood floors"

NEGATIVE (outdated / poor condition)

"an outdated home with old carpet, worn surfaces and dated wallpaper"

"a house needing renovation with peeling paint, cracked walls and old fixtures"

4 positive + 4 negative prompts total

Evaluation Results — Structured Model

8,445 held-out test homes · 54 states · structured-only (zero API cost)

42.1%
Mean Absolute % Error (MAPE)
overall

28.3%
Median Absolute % Error
more representative

0.610
RMSE (log-price)
vs 0.494 on validation

MAPE by Price Tier

Price Tier	MAPE	Median APE	n	Note
$0–300k	69.2%	36.8%	2,148	Distressed/foreclosure sales skew this tier
$300–600k	28.8%	21.1%	2,145	Best tier — densest training data
$600k–1M	31.0%	24.6%	2,105	AVM + market signal closes the gap at runtime
$1M+	39.2%	35.4%	2,047	Thin comps — premium boost helps at inference

Best: Arizona 28.8% · Kansas 31.1% · Illinois 31.4% · Maryland 32.6%

Highest: California 54.5% · Hawaii 54.5% · Florida 51.8% · DC 46.4%

Real-World Validation & Conclusion

Full-Pipeline Test — Known Sold Home

Property:	Olney, MD · 4 bd / 3 ba / 2,552 sqft
Actual:	$931,000
Predicted:	**$929,592**
Error:	**0.15%**
Signals:	Structured + AVM + Market + CLIP (3 photos)
Mechanism:	Outlier detection fired; premium boost applied

Structured-only: ~$370,000 (-60% error) — full pipeline is essential for premium markets

Key Takeaways

- Structured XGBoost: solid mid-market baseline (28% median APE)
- AVM blend reduces premium market error from ~70% to ~10%
- CLIP condition scoring adds meaningful ±10–15% visual adjustment
- Outlier detection is critical for premium non-CA markets
- Full pipeline: <1% error on tested premium sold home ($931k)

Limitations & Future Work

⚠ Known Limitations

- $0–300k tier MAPE 69%: distressed sales and data quality skew this segment
- CLIP calibration $R^2 < 0$: only 89 training images — centered fallback in use
- No Zillow AVM: API returns wrong-state results, removed from production
- CA homes: model underpredicts (multi-state training reduced CA-specific learning)

>> Future Improvements

- Filter training floor to $50k+ to remove distressed sales from $0–300k tier
- Collect 200+ more images for CLIP calibration (411 API calls remaining)
- Add latitude/longitude features for finer geographic resolution within cities
- State-specific XGBoost fine-tuning for CA, FL, HI outlier states

Multi-signal stacking transforms a 42% MAPE baseline into a sub-1% full-pipeline result on known premium homes — demonstrating the value of layered ML architecture.